

issues

Issues

Revision: 241 **Last modified:** 2026-06-16T23:42:27Z **Description:** Active workable-item registry for Helix Cluster OS **Authority:** Constitution §11.4.93 (workable-items DB single source of truth) **Generated-by:** scripts/docs/db_to_md.py (DB is canonical; edit via cmd/hxc-registry, not by hand)

Total active items: **223**. Canonical source: data/hxc_registry.db. Full per-item detail (description, closure criteria, required test types, source refs) lives in the DB and docs/research/_ledger/*.json.

Foundation (anti-bluff) (7 active)

HXC	Type	Pri	Status	Title
HXC-932	Task	P0	Queued	Wire-in vs prune decision + execution for ~167 orphaned pkg packages (STRATEGIC umbrella)
HXC-016	Task	P1	Queued	
HXC-017	Task	P1	Queued	
HXC-018	Task	P1	Queued	
HXC-019	Task	P1	Queued	
HXC-020	Task	P1	Queued	
HXC-949	Feature	P2	Queued	Chaos OTel polyglot trace (Rust/Elixir/Go) + Grafana dashboards (split from HXC-1258)

MVP (5 active)

HXC	Type	Pri	Status	Title
HXC-1100	Feature	P1	Queued	Implement real Bazel RBE build execution (internal/build)
HXC-1140	Task	P1	Queued	Publish v1.0.0-dev-mvp release across all modules with packaging
HXC-1128	Feature	P2	Queued	Implement Health Monitor with eBPF probes and LSTM failure prediction
HXC-1129	Feature	P2	Queued	Implement CRIU/DMTCP-based session live migration orchestration
HXC-1143	Feature	P2	Queued	Implement distcc/icecream distributed C/C++ compilation and ccache layer

Phase 2 (9 active)

HXC	Type	Pri	Status	Title
HXC-1153	Feature	P1	Queued	Implement pkg/resources Linux GPU reader via /sys/class/drm (NVML-free)
HXC-1159	Task	P1	Queued	E2E: a Linux node joins the cluster mesh and becomes schedulable
HXC-906	Docs	P1	Queued	PHASE_2_ROADMAP.md
HXC-1154	Feature	P2	Queued	

HXC	Type	Pri	Status	Title
				Integrate universal Vulkan compute backend (vendor-neutral, no device-specific code)
HXC-1155	Feature	P2	Queued	Implement console/node thermal sink reader via /sys/class/thermal
HXC-1161	Feature	P2	Queued	E2E: aggregate GPU pool runs llama.cpp inference across heterogeneous nodes
HXC-1163	Task	P2	Queued	Validate cross-NAT WireGuard mesh hole-punching across real hosts
HXC-1166	Research	P3	Queued	CRIU/DMTCP live process migration (research + spike)
HXC-931	Task	P3	Queued	Reconcile internal/console linux_boot.go (BootMachine) with boot_coordinator.go (BootCoordinator)

Phase 3 (31 active)

HXC	Type	Pri	Status	Title
HXC-1168	Feature	P0	Queued	Implement SBC Linux node agent (Orange Pi 5 Max target, Tier T3)
HXC-1171	Task	P0	Queued	Validate Mali-G610 Vulkan/OpenCL compute on SBC
HXC-1172	Feature	P0	Queued	Scaffold Android Agent APK (Kotlin + NDK + Termux)
HXC-1173	Feature	P0	Queued	Implement Android foreground service framework with START_STICKY
HXC-1174	Feature	P0	Queued	

HXC	Type	Pri	Status	Title
				Integrate Android BatteryManager monitoring
HXC-1205	Feature	P0	Queued	Build edge setup wizard for device onboarding
HXC-1208	Task	P0	Queued	Implement battery/thermal stress testing for edge devices
HXC-1169	Feature	P1	Queued	Implement SBCAdapter for SBC-specific hardware monitoring
HXC-1170	Feature	P1	Queued	Integrate RK3588 NPU via RKNN Toolkit2 C API
HXC-1176	Feature	P1	Queued	Implement Android Vulkan/ NNAPI compute backend
HXC-1177	Feature	P1	Queued	Implement QUIC transport client for Android (Tier T6)
HXC-1179	Feature	P1	Queued	Scaffold iOS Agent app (Swift, Tier T7 EDGE_DONOR)
HXC-1180	Feature	P1	Queued	Implement iOS Metal compute integration
HXC-1182	Feature	P1	Queued	Implement iOS background execution scheduling (BGAppRefresh/ BGProcessing)
HXC-1192	Feature	P1	Queued	Implement unified NPU backend with ONNX model converter
HXC-1194	Feature	P1	Queued	Integrate MLC LLM universal engine for mobile inference

HXC	Type	Pri	Status	Title
HXC-1200	Feature	P1	Queued	Implement Jetson CUDA-on-ARM GPU backend
HXC-1203	Task	P1	Queued	Implement Armbian TV-box provisioning path (RK3588/ Amlogic)
HXC-1204	Task	P1	Queued	Implement ADB- over-WiFi fleet provisioning for Android cluster
HXC-1206	Feature	P1	Queued	Build APK/IPA distribution system
HXC-1209	Task	P1	Queued	Validate ARM SBC per-watt performance >=80% of x86
HXC-926	Task	P1	Queued	Wire pkg/ powergater CanAcceptWork into edge/agent work-acceptance path
HXC-1181	Feature	P2	Queued	Implement iOS CoreML / Neural Engine inference engine
HXC-1183	Feature	P2	Queued	Implement HarmonyOS Agent (ArkTS) with Da Vinci NPU inference
HXC-1217	Feature	P2	Queued	Implement internal/gpu cgo vendor backends (CUDA/ROCm/ oneAPI/MLX)
HXC-1218	Feature	P2	Queued	Implement session live migration (CRIU/ DMTCP)
HXC-1219	Feature	P2	Queued	Implement WireGuard UPnP/NAT-PMP NAT traversal
HXC-1221	Docs	P2	Queued	Document Phase 3 edge multi-

HXC	Type	Pri	Status	Title
				platform agent architecture diagrams
HXC-1622	Feature	P2	Queued	HXC-1167 follow-up: CGO-enabled aarch64 cross-build via zig cc / aarch64-linux-gnu-gcc
HXC-1623	Task	P2	Queued	HXC-1167 follow-up: validate cross-compiled helix-agent on real Orange Pi 5 Max (aarch64) execution
HXC-1624	Feature	P2	Queued	HXC-1158 follow-up: add positive non-console SoC-match labels (RK3588/RPi/x86 server) to internal/console detector

Phase 4 (44 active)

HXC	Type	Pri	Status	Title
HXC-1222	Feature	P0	Queued	Build K3s orchestration substrate with RuntimeClasses for firecracker/kata/runc
HXC-1223	Feature	P0	Queued	Implement Firecracker microVM provisioner for T1-T3 with vsock and virtio-net
HXC-1224	Feature	P0	Queued	Implement Firecracker golden snapshot create/restore pipeline (<=28ms restore)
HXC-1232	Feature	P0	Queued	Implement unified golden snapshot + COW instant-reset across all simulator backends
HXC-1234	Feature	P0	Queued	Build Rust DST engine: single-threaded SimLoop

HXC	Type	Pri	Status	Title
				with virtual clock and seeded PRNG
HXC-1235	Feature	P0	Queued	Integrate Rust turmoil for deterministic simulated TCP/UDP networking
HXC-1242	Feature	P0	Queued	Build Elixir/OTP Chaos Controller GenServer with supervision-tree isolation
HXC-1251	Feature	P0	Queued	Build Virtual Testing Controller OTP supervision tree (one_for_all)
HXC-1252	Feature	P0	Queued	Implement SessionManager with 50-session cap, 2h TTL, resource-quota enforcement
HXC-1253	Feature	P0	Queued	Implement DevicePool GenServer with tier-to-simulator dispatch and health checks
HXC-1256	Feature	P0	Queued	Expose REST API for session CRUD, device provisioning and snapshot restore
HXC-1225	Feature	P1	Queued	Implement QEMU/KVM ARM64 virt provisioner for T6 RK3588-approximate SBC
HXC-1226	Feature	P1	Queued	Integrate Cuttlefish/CrosVM provisioner for T5 Android AOSP devices
HXC-1228	Feature	P1	Queued	Implement Docker + binfmt_misc protocol stubs for T7 iOS and T8 HarmonyOS
HXC-1250	Feature	P1	Queued	Integrate Chaos Mesh CRDs (NetworkChaos, TimeChaos, StressChaos, DNSChaos)
HXC-1257	Feature	P1	Queued	Build Phoenix LiveView real-time test dashboard
HXC-1262	Feature	P1	Queued	Integrate CI/CD quality gates for GitHub Actions / GitLab CI / Jenkins
HXC-1263	Feature	P1	In progress	Build WebAssembly plugin host on Wasmtime Component Model with WIT bindings
HXC-1269	Feature	P1	Queued	Build cmd/helix-testd daemon as OTP controller node entryptpoint

HXC	Type	Pri	Status	Title
HXC-1273	Feature	P1	Queued	Implement vsock host-guest control channel for chaos-resilient telemetry
HXC-1284	Task	P1	Queued	Benchmark Firecracker 5,000+ microVMs per host with KSM and memory overcommit
HXC-1285	Task	P1	Queued	Run 72-hour continuous chaos soak proving no resource leaks or degradation
HXC-1286	Docs	P1	Queued	Execute Production Readiness Review with 80-item checklist ($\geq 95\%$ complete)
HXC-1287	Task	P1	Queued	Run full 8-tier matrix (160 nodes) end-to-end within 45-minute CI budget
HXC-907	Docs	P1	Queued	PHASE_4_ROADMAP.md
HXC-1227	Feature	P2	Queued	Implement T4 gaming-console protocol-level constrained x86_64 VM
HXC-1246	Feature	P2	Queued	Implement 6 hardware fault injectors (NMI, mem CE/UE, PCIe AER, CPU bit-flip, thermal throttle)
HXC-1266	Docs	P2	Queued	Provide reference plugins (Rust device-sim, Zig device-sim) compiled to wasm32-wasi
HXC-1270	Feature	P2	Queued	Implement Firecracker MicroVM Kubernetes operator and CRD
HXC-1271	Feature	P2	Queued	Deploy WireGuard encrypted inter-host mesh DaemonSet for multi-host clusters
HXC-1272	Feature	P2	Queued	Implement distributed snapshot pool backend (MinIO) and multi-host scheduling
HXC-1274	Feature	P2	Queued	Implement Mininet WAN topology harness for multi-region network-stack testing
HXC-1275	Research	P2	Queued	Author TLA+/PlusCal formal specs for leader election and scheduler allocation
HXC-1276	Research	P2	Queued	

HXC	Type	Pri	Status	Title
				Author Jepsen black-box linearizability test for task scheduling under partitions
HXC-1279	Feature	P2	Queued	Add eBPF/XDP line-rate programmable network fault injection (cilium/ebpf)
HXC-1283	Docs	P2	Queued	Define Corellium / physical-hardware-in-the-loop path for iOS, PS4, RK3588 GPU/NPU
HXC-1288	Task	P2	Queued	Define polyglot interop boundaries (Rust↔Go↔Elixir↔Wasm)
HXC-1289	Feature	P2	Queued	Integrate libcluster automatic BEAM cluster formation via Kubernetes DNS
HXC-1625	Bug	P2	Queued	Replicate SELinux relabel fix (HXC-1621) to Herald/containers/pkg/crossbuild duplicate copy
HXC-921	Feature	P2	Queued	Bazel Remote Execution (REAPI) interop for build service (optional)
HXC-1277	Research	P3	Queued	Integrate Shadow/Phantom for unmodified-binary deterministic network simulation
HXC-1280	Research	P3	Queued	Validate gem5 big.LITTLE (O3+Minor) scheduler fidelity for RK3588
HXC-1281	Research	P3	Queued	Evaluate VirGL/Venus virtual GPU for OpenGL/Vulkan workload functional testing
HXC-1282	Research	P3	Queued	Integrate Tart OCI-native macOS/Linux VMs for iOS/macOS CI build-and-test

Phase 5 (26 active)

HXC	Type	Pri	Status	Title
HXC-1314	Feature	P0	Queued	Implement pkg/handheld Steam Deck agent (Vulkan compute, battery, thermal) (DEFERRED hw)

HXC	Type	Pri	Status	Title
HXC-1312	Feature	P1	Queued	Build multi-arch universal agent container (amd64/arm64/riscv64) via buildx + distroless
HXC-1313	Feature	P1	Queued	Implement riscv64 cross-compilation pipeline + RVV extension detection (DEFERRED CI)
HXC-1315	Feature	P1	Queued	Package Steam Deck Flatpak agent with systemd integration (DEFERRED hw)
HXC-1316	Feature	P1	Queued	Implement internal/sbc: Jetson TensorRT backend (≥ 60 TOPS) (DEFERRED hw)
HXC-1317	Feature	P1	Queued	Implement internal/sbc RK3588 NPU + Turing RK1 cluster support
HXC-1320	Feature	P1	Queued	Implement internal/enterprise EPYC/Ampere auto-provisioning + Coreboot trust detection (DEFERRED hw)
HXC-1321	Feature	P1	Queued	Implement internal/iot OpenWrt router gateway agent (GL-MT6000, Docker, $< 5\%$ overhead)
HXC-1322	Feature	P1	Queued	Implement internal/iot NAS storage-node agent (Synology/QNAP Docker, dual-role)
HXC-1327	Feature	P1	Queued	Implement gVisor/Kata per-tier sandbox hardening for UNTRUSTED/edge nodes (DEFERRED)
HXC-1329	Feature	P1	Queued	Implement hybrid cloud-on-prem WireGuard mesh

HXC	Type	Pri	Status	Title
HXC-1332	Task	P1	Queued	joining spot workers to on-prem cluster Phase 5a integration test: 10-node mixed handheld+SBC cluster
HXC-1318	Feature	P2	Queued	Implement pkg/fpga + internal/fpga hard-SoC/soft-core/DPU backends (DEFERRED hw)
HXC-1319	Feature	P2	Queued	Implement bitstream verification + management for FPGA tiers (T11 required)
HXC-1324	Feature	P2	Queued	Implement internal/exotic Groq LPU inference adapter (<100ms TTFT) (DEFERRED cloud)
HXC-1325	Feature	P2	Queued	Implement internal/exotic Cerebras CS-3 large-model inference adapter (DEFERRED cloud)
HXC-1333	Task	P2	Queued	Phase 5b integration test: 8-node arm64+riscv64+fpga heterogeneous cluster
HXC-1617	Task	P2	Queued	Live-cloud IMDS interruption integration on real AWS/Azure/GCP spot accounts
HXC-1618	Task	P2	Queued	Validate pkg/benchmark normalized scores across two physically different nodes
HXC-928	Task	P2	Queued	Execute linux fixtures on real linux CI for wave-74 packages (CLAUDE-2)
HXC-929	Task	P2	Queued	Real macOS thermal source for

HXC	Type	Pri	Status	Title
				powergater/ edgeheartbeat CPU temp
HXC-930	Task	P2	Queued	Live discrete-GPU/ NPU hardware probe for pkg/ resources accelerators
HXC-1323	Feature	P3	Queued	Implement webOS smart-TV JS agent over WebSocket (DEFERRED device)
HXC-1326	Research	P3	Queued	Implement pkg/ quantum Qiskit Runtime plugin for async circuit submission (DEFERRED)
HXC-1330	Docs	P3	Queued	Define and validate 5 reference cluster build recipes (\$250/\$500/\$1k/ \$2k/\$5k)
HXC-1620	Bug	P3	Queued	Harden tierdetect linux cpuinfoIsARM64 multi-digit CPU- architecture parsing

Phase 6 (17 active)

HXC	Type	Pri	Status	Title
HXC-1369	Feature	P0	Queued	Implement cmd/helix- federation binary and helixctl federation CLI
HXC-1384	Task	P0	Queued	Establish multi-cell integration testbed for CLAUDE-1 usability gate
HXC-933	Feature	P0	Queued	Build cmd/ helix- federation and

HXC	Type	Pri	Status	Title
				wire Phase 6 packages (federation/crdt/hlc/spiffefed/gitops)
HXC-1362	Feature	P1	Queued	Implement pkg/cilium Cluster Mesh client and global-service propagation
HXC-1370	Feature	P1	Queued	Extend internal/node cell agent: gossip-tier selection + federation identity attestation
HXC-1385	Task	P1	Queued	Validate Phase 6 exit-gate KPIs against the live federation
HXC-1354	Feature	P2	Queued	Implement SSH tunnel bridge fallback for control-plane bootstrap
HXC-1355	Feature	P2	Queued	Implement cloud VPN bridge with provider-specific endpoint discovery
HXC-1360	Feature	P2	Queued	Integrate External Secrets Operator + Vault for federation secrets
HXC-1366	Feature	P2	Queued	Implement cloud bursting auto-scaling burst pool with spot

HXC	Type	Pri	Status	Title
				termination handling
HXC-1378	Feature	P2	Queued	Implement hierarchical Prometheus federation across cells
HXC-1379	Feature	P2	Queued	Implement OpenTelemetry cross-cell distributed tracing
HXC-1382	Feature	P2	Queued	Implement Velero tiered disaster recovery backup and restore
HXC-922	Task	P2	Queued	Live ArgoCD 3-cell ApplicationSet rollout usability capture (HXC-1367 live gate)
HXC-924	Task	P2	Queued	Live Karmada multi-cluster <60s failover usability capture (HXC-1364 live gate)
HXC-925	Task	P2	Queued	Two-node LAN mDNS discovery usability capture (HXC-1347 live gate)
HXC-1376	Docs	P3	Queued	Establish quarterly Game Day federation chaos drill protocol

Phase 7 (4 active)

HXC	Type	Pri	Status	Title
HXC-1414	Feature	P2	Queued	Implement NATS leaf-node edge-to-core store-and-forward topology (G-12, P2)
HXC-1619	Task	P2	Queued	Enable cmd/dst-sim 1000-seed gate as a live CI job
HXC-1423	Docs	P3	Queued	Author TLA+ formal specifications for consensus/coordination protocols (P2)
HXC-914	Task	P3	Queued	internal/chaos: author live-cluster nightly pipeline + GitHub Actions workflow

Phase 8 (40 active)

HXC	Type	Pri	Status	Title
HXC-1434	Feature	P0	Queued	Implement GraVal-style miner attestation handshake controller in pkg/chutes
HXC-1438	Feature	P0	Queued	Implement pkg/chutes MinerController K3s deployment lifecycle
HXC-1440	Feature	P0	Queued	Deploy PostgreSQL StatefulSet for miner inventory tracking

HXC	Type	Pri	Status	Title
HXC-1441	Feature	P0	Queued	Deploy Redis pub/sub Deployment for miner event propagation
HXC-1442	Feature	P0	Queued	Deploy GraVal bootstrap privileged DaemonSet for GPU attestation
HXC-1443	Feature	P0	Queued	Deploy miner-api Deployment with NodePort 32000
HXC-1444	Feature	P0	Queued	Deploy Gepetto strategy engine via ConfigMap-backed Deployment
HXC-1445	Feature	P0	Queued	Deploy registry-proxy host-network DaemonSet on port 30500
HXC-1450	Feature	P0	Queued	Implement pkg/bittensor wallet (coldkey/hotkey) and TAO balance
HXC-1451	Feature	P0	Queued	Implement Bittensor SN64 subnet registration flow
HXC-1462	Feature	P0	Queued	Author helm/helixcluster-chutes base chart (values.yaml)
HXC-1465	Feature	P0	Queued	Author scripts/chutes-node-prep.sh bare-metal node preparation
HXC-1466	Feature	P0	Queued	Author scripts/chutes-miner-deploy.sh miner deployment + registration
HXC-1491	Task	P0	Queued	Validate end-to-end full request flow integration

HXC	Type	Pri	Status	Title
				(workload->client->E2EE->network->miner->response)
HXC-1463	Feature	P1	Queued	Author chute-deployment.yaml Helm template for inference pods
HXC-1464	Feature	P1	Queued	Author values-models.yaml with 8 pre-configured model deployments
HXC-1467	Feature	P1	Queued	Author scripts/chutes-health-monitor.sh observability + alerts
HXC-1468	Feature	P1	Queued	Author scripts/chutes-verify.sh end-to-end verification
HXC-1471	Feature	P1	Queued	Deploy vLLM inference engine with PagedAttention defaults
HXC-1472	Feature	P1	Queued	Deploy SGLang inference engine with RadixAttention
HXC-1485	Task	P1	Queued	Author HelixQA Challenge: miner deployment and TAO earnings
HXC-1486	Task	P1	Queued	Author HelixQA Challenge: E2EE inference roundtrip
HXC-1487	Task	P1	Queued	Author HelixQA Challenge: multi-marketplace routing decision
HXC-1488	Task	P1	Queued	Author HelixQA Challenge:

HXC	Type	Pri	Status	Title
				GraVal attestation pass/ fail
HXC-1489	Task	P1	Queued	Validate Chutes API client P99 first-token streaming latency < 500ms
HXC-935	Feature	P1	Queued	Wire Phase 8 marketplace/ GraVal/miner into a real control loop
HXC-936	Bug	P1	Queued	Resolve dual package trees: top-level pkg/ gputest vs security/pkg/ gputest (and similar)
HXC-1432	Feature	P2	Queued	Implement gzip compression pipeline in E2EE inference envelope
HXC-1433	Task	P2	Queued	Reconcile E2EE AEAD spec deviation (AES-256-GCM vs roadmap ChaCha20- Poly1305)
HXC-1437	Research	P2	Queued	Implement GraVal CUDA Proof-of- Consecutive- VRAM-Work kernel (DEFERRED, GPU)
HXC-1452	Feature	P2	Queued	Implement Yuma Consensus / metagraph weight queries in pkg/bittensor
HXC-1457	Feature	P2	Queued	Implement io.net marketplace adapter
HXC-1459	Feature	P2	Queued	

HXC	Type	Pri	Status	Title
				Implement Salad marketplace adapter
HXC-1475	Feature	P2	Queued	Deploy Intel TDX + NVIDIA CC TEE stack via sek8s
HXC-1477	Feature	P2	Queued	Implement Cosign/Sigstore image admission controller for K3s
HXC-1490	Task	P2	Queued	Achieve >60% line coverage on pkg/chutes, pkg/marketplace, pkg/e2ee (Codecov gate)
HXC-1453	Feature	P3	Queued	Implement child-hotkey delegation support in pkg/bittensor
HXC-1474	Feature	P3	Queued	Deploy TurboDiffusion video-generation engine
HXC-1478	Feature	P3	Queued	Implement LUKS-encrypted root for miner nodes
HXC-1479	Feature	P3	Queued	Implement Cilium egress control and Watchtower integrity challenges

Phase 8B (28 active)

HXC	Type	Pri	Status	Title
HXC-1492	Feature	P0	Queued	Define pkg/pool GPUProvider interface as single integration seam
HXC-1521	Task	P0	Queued	Expose security ML-KEM-768 E2EE Session/

HXC	Type	Pri	Status	Title
				Transport as pkg/ e2ee
HXC-1522	Feature	P0	Queued	Implement pkg/ e2ee E2EProxy outbound remote- GPU traffic proxy
HXC-1552	Task	P0	Queued	Implement HelixQA BurstChallenge saturate+failover exit gate
HXC-1553	Task	P0	Queued	Implement E2E helix submit -> saturate -> route- to-Chutes demo
HXC-1520	Feature	P1	Queued	Implement LocalGPURegistrar nvidia-smi discovery + GPU classification
HXC-1526	Feature	P1	Queued	Implement pkg/ proxy CUDA API interceptor + memory staging
HXC-1528	Feature	P1	Queued	Implement virtual / dev/nvidia* device creation proxying to remote GPU
HXC-1536	Feature	P1	Queued	Register virtual GPU devices into internal/node cluster resource model
HXC-1537	Task	P1	Queued	Integrate GraValVerifier into pkg/security provider admission
HXC-1540	Feature	P1	Queued	Implement local vLLM + SGLang serving stack deployment
HXC-1541	Feature	P1	Queued	Implement GPU- proxy DaemonSet advertising virtual nvidia.com/gpu
HXC-1549	Task	P1	Queued	Author Helm chart for Phase 8B GPU pool + burst deployment

HXC	Type	Pri	Status	Title
HXC-1554	Task	P1	Queued	Stand up real-provider CI integration (Chutes/io.net/RunPod)
HXC-1555	Task	P1	Queued	Capture E2EE usability evidence: tee=true encrypted-traffic gate
HXC-1557	Task	P1	Queued	Run inference p99 <500ms load test via Chutes (1000 requests)
HXC-1615	Feature	P1	Queued	Wire pkg/gpu ProviderAdapter registry into pkg/pool provider consumption
HXC-1517	Feature	P2	Queued	Implement AWS Spot 2-minute interrupt -> CRIU/S3 checkpoint failover
HXC-1524	Feature	P2	Queued	Implement pkg/e2ee TDX AttestationVerifier for sensitive workloads
HXC-1525	Feature	P2	Queued	Implement key rotation CronJob + HSM integration for E2EE keys
HXC-1543	Feature	P2	Queued	Implement KEDA queue-depth/custom-metric autoscaling
HXC-1545	Feature	P2	Queued	Implement CRIU GPU-context checkpoint/restore + graceful degradation
HXC-1550	Task	P2	Queued	Author Docker Compose development stack for Phase 8B
HXC-1558	Task	P2	Queued	Run 48-hour load test and 7-day soak (>99.9% success)
HXC-1559	Task	P2	Queued	

HXC	Type	Pri	Status	Title
				Run chaos suite for provider failover and DDoS resilience
HXC-1518	Docs	P3	Queued	Document GCP and Azure provider adapter patterns
HXC-1542	Feature	P3	Queued	Implement bittencert blockchain-backed X.509 identity
HXC-1546	Feature	P3	Queued	Implement chutes-dropzone self-hosted gateway integration

Phase 8C (12 active)

HXC	Type	Pri	Status	Title
HXC-1560	Feature	P0	Queued	Implement pkg/security/e2ee PQ envelope (ML-KEM-768 + HKDF-SHA256)
HXC-1562	Task	P0	Queued	Add Chutes byte-interop test-vector corpus for pkg/security/e2ee
HXC-1574	Feature	P0	Queued	Implement pkg/security/ attestation TEE quote construct/verify
HXC-1575	Feature	P0	Queued	Implement Intel TDX quote generation with nonce+cert binding (report_data layout)
HXC-1576	Feature	P0	Queued	Implement NVIDIA GPU attestation evidence verification (NRAS/nvTrust)
HXC-1566	Feature	P1	Queued	Implement E2EE instance-

HXC	Type	Pri	Status	Title
				discovery + single-use nonce contract
HXC-1577	Feature	P1	Queued	Implement measured-boot / RTMR3 access-config measurement
HXC-1578	Feature	P1	Queued	Implement attestation-gated key release (LUKS/secret) handshake
HXC-1591	Feature	P1	Queued	Extend pkg/resources GPUInfo with device fingerprint + attested/thermal state
HXC-1594	Feature	P1	Queued	Implement LLMOrchestrator vLLM/SGLang backend WRAP (OpenAI HTTP)
HXC-1593	Feature	P2	Queued	Implement VRAM-capacity (>=95%) usable-memory gate in resources/gpuattest
HXC-1611	Research	P3	Queued	Implement Karmada-pattern multi-cluster aggregation over SWIM+Raft+etcd (Phase 6 hook)